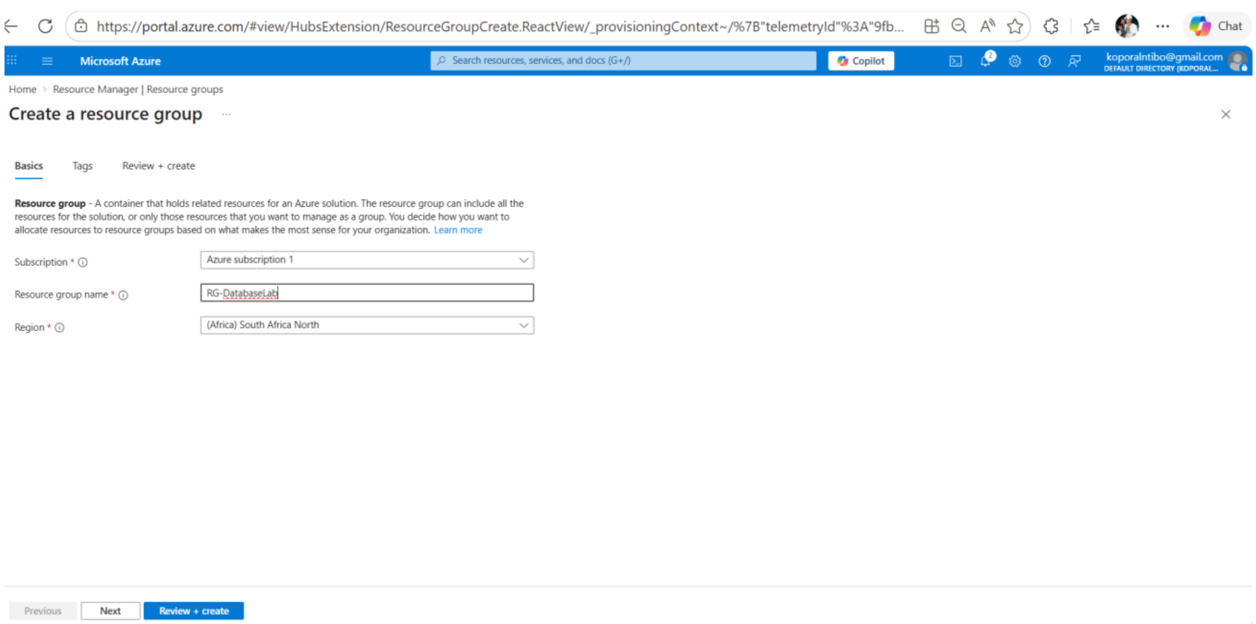
CHAPTER 05 LAB

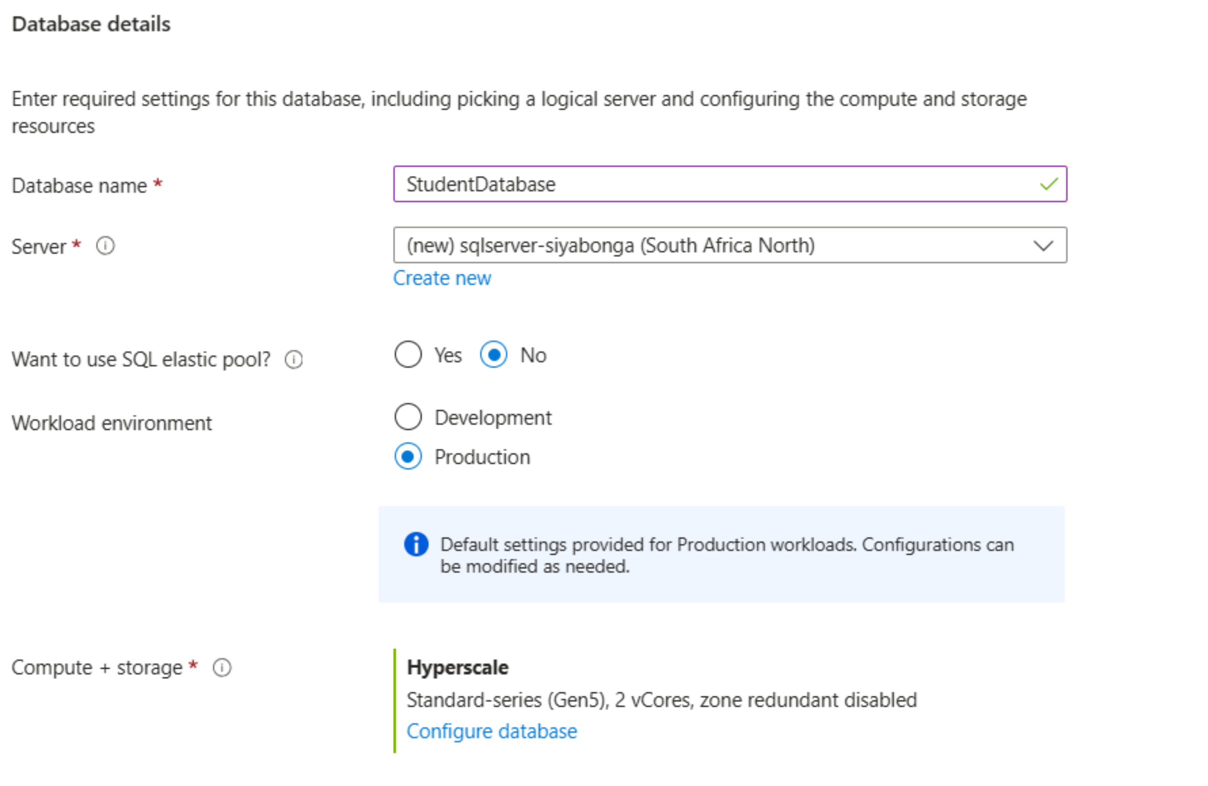
AZURE DATABASE SERVICES

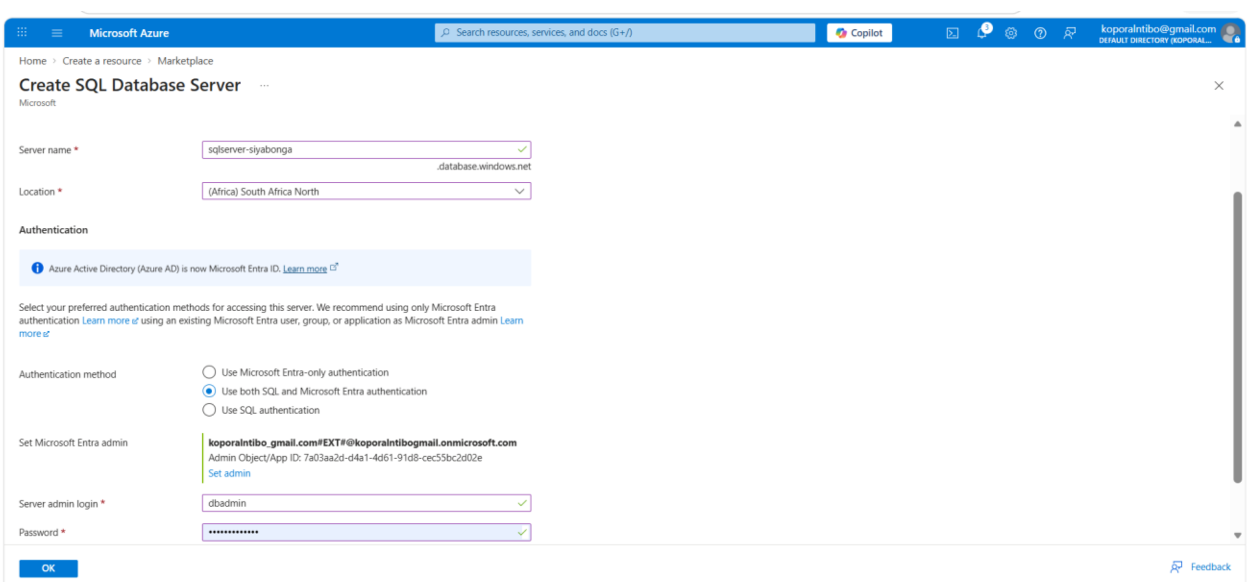
LAB 5: Implementing and Managing Azure Databases

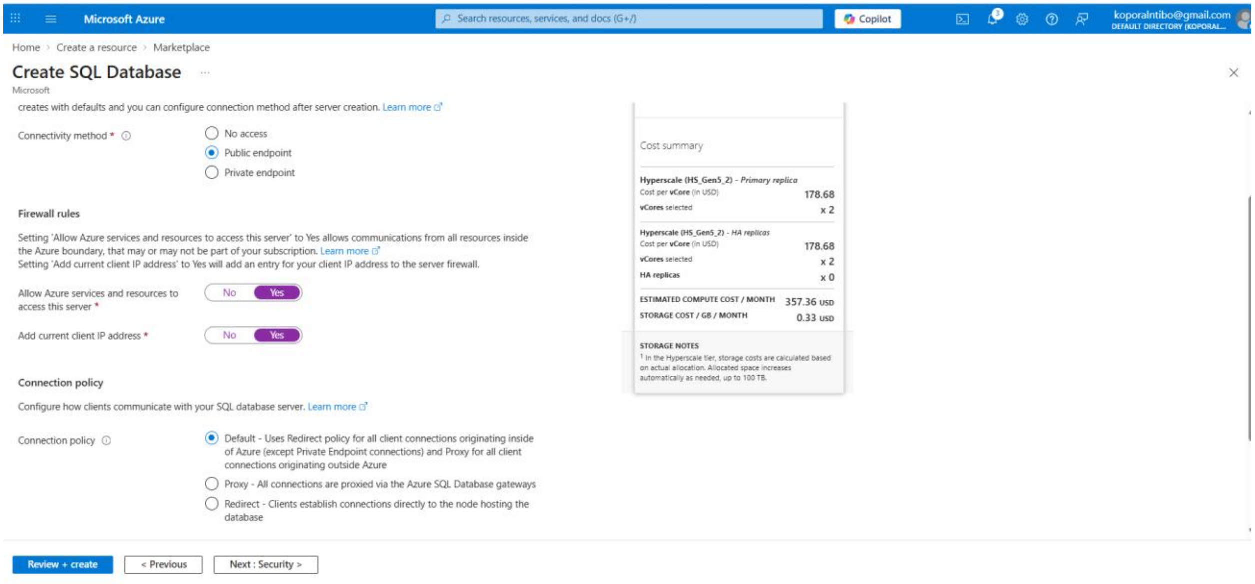
TASK 1: CREATE AZURE SQL DATABASE

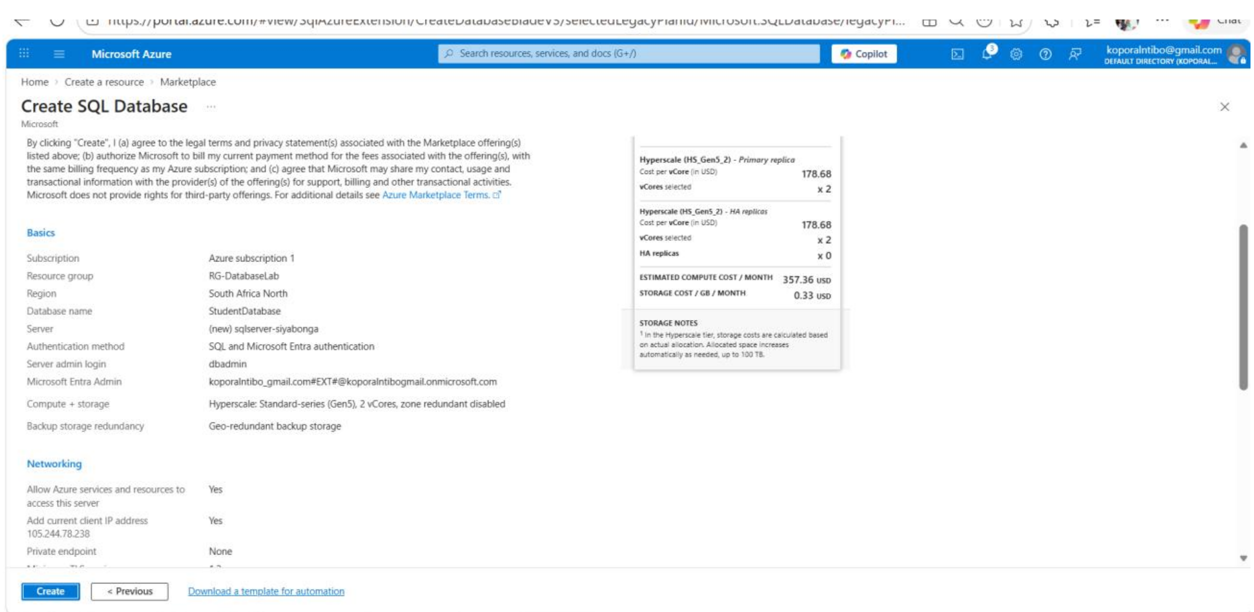
I chose Azure subscription 1 for this lab because it provides access to the necessary Azure services under my student account, allowing me to create and manage resources such as virtual machines and databases without additional cost. This subscription supports the deployment of cloud infrastructure required for the exercise, making it suitable for building and testing solutions within the Azure environment.




Database details
Enter required settings for this database, including picking a logical server and configuring the compute and storage resources
Database name *
StudentDatabase
Server *
(new) sqlserver-siyabonga (South Africa North)
Create new
Want to use SQL elastic pool?
Yes
No
Workload environment
Development
Production
Default settings provided for Production workloads. Configurations can be modified as needed.
Compute + storage *
Hyperscale
Standard-series (Gen5), 2 vCores, zone redundant disabled
Configure database

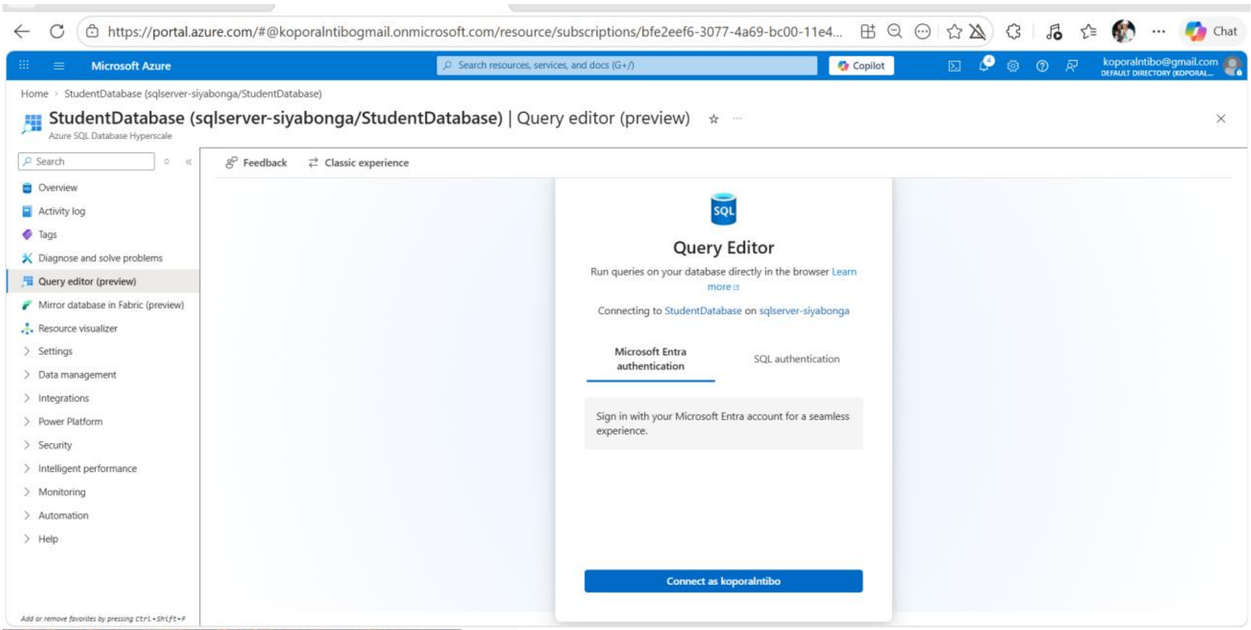

Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
3
koporalntibo@gmail.com
DEFAULT DIRECTORY (KOPORAL...
Home › Create a resource › Marketplace
Create SQL Database Server
Microsoft
Server name *
sqlserver-siyabonga
.database.windows.net
Location *
(Africa) South Africa North
Authentication
Azure Active Directory (Azure AD) is now Microsoft Entra ID. Learn more
Select your preferred authentication methods for accessing this server. We recommend using only Microsoft Entra authentication Learn more using an existing Microsoft Entra user, group, or application as Microsoft Entra admin Learn more
Authentication method
Use Microsoft Entra-only authentication
Use both SQL and Microsoft Entra authentication
Use SQL authentication
Set Microsoft Entra admin
koporalntibo_gmail.com#EXT#@koporalntibogmail.onmicrosoft.com
Admin Object/App ID: 7a03aa2d-d4a1-4d61-91d8-cec55bc2d02e
Set admin
Server admin login *
dbadmin
Password *
OK
Feedback



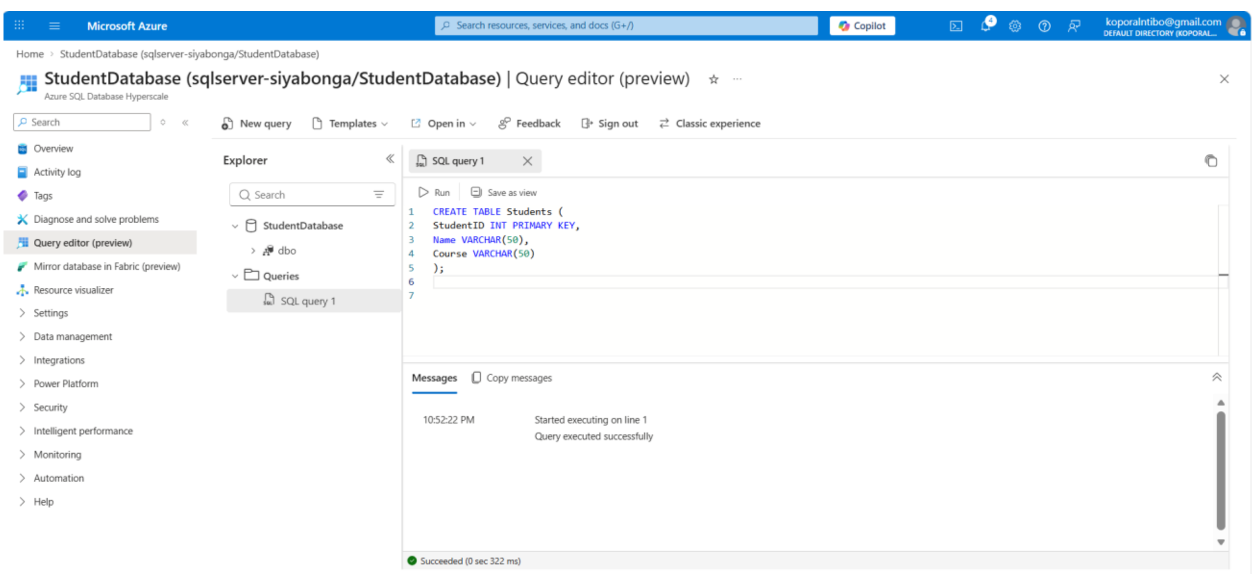


Task 2: Create Database Table

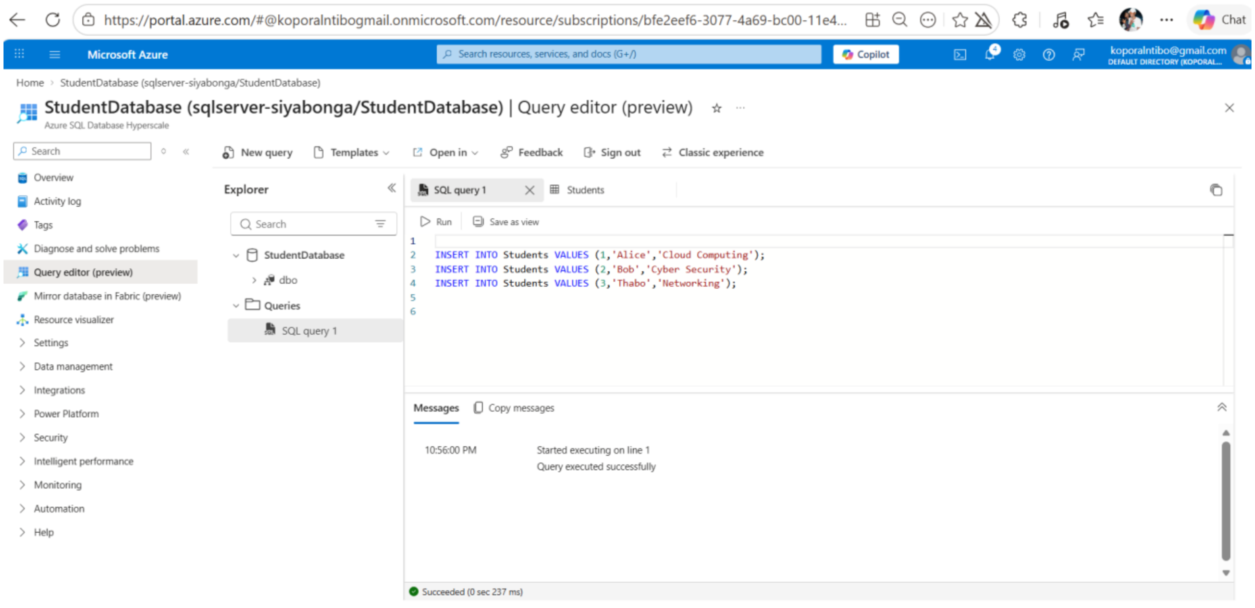
After the server is created and deployed, enter the Query Editor tab and select the SQL authentication option. Once the SQL authentication option is selected, use the server authentication credentials to connect to the server.

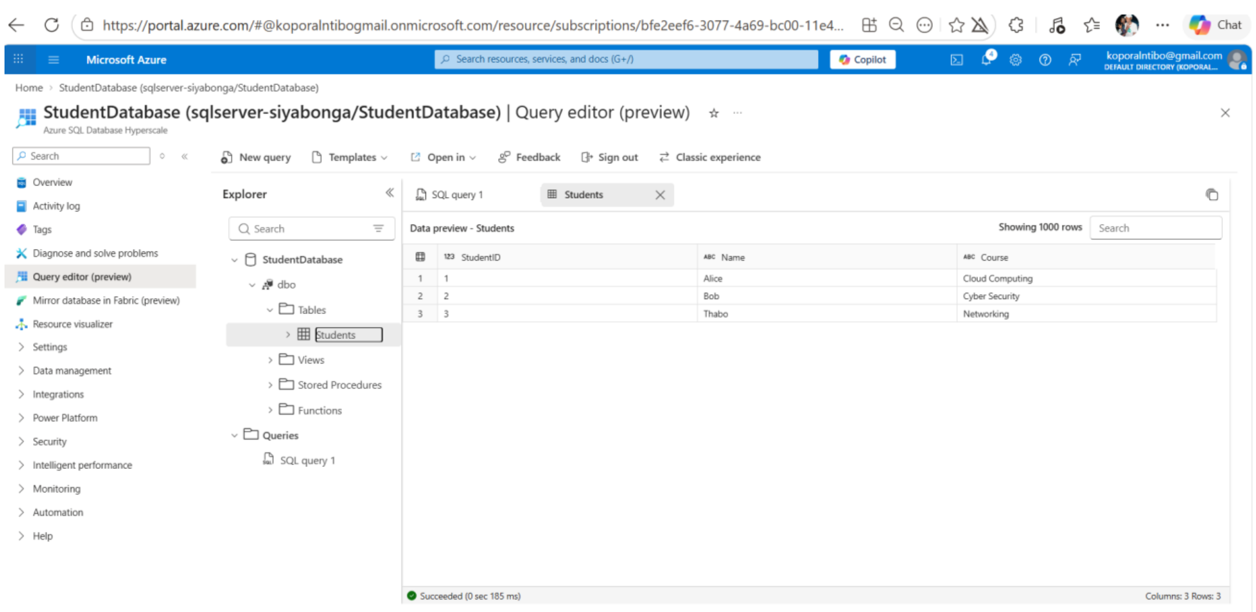


Once connected to the server, enter the SQL commands provided to create a SQL table structure in the server, then run it.



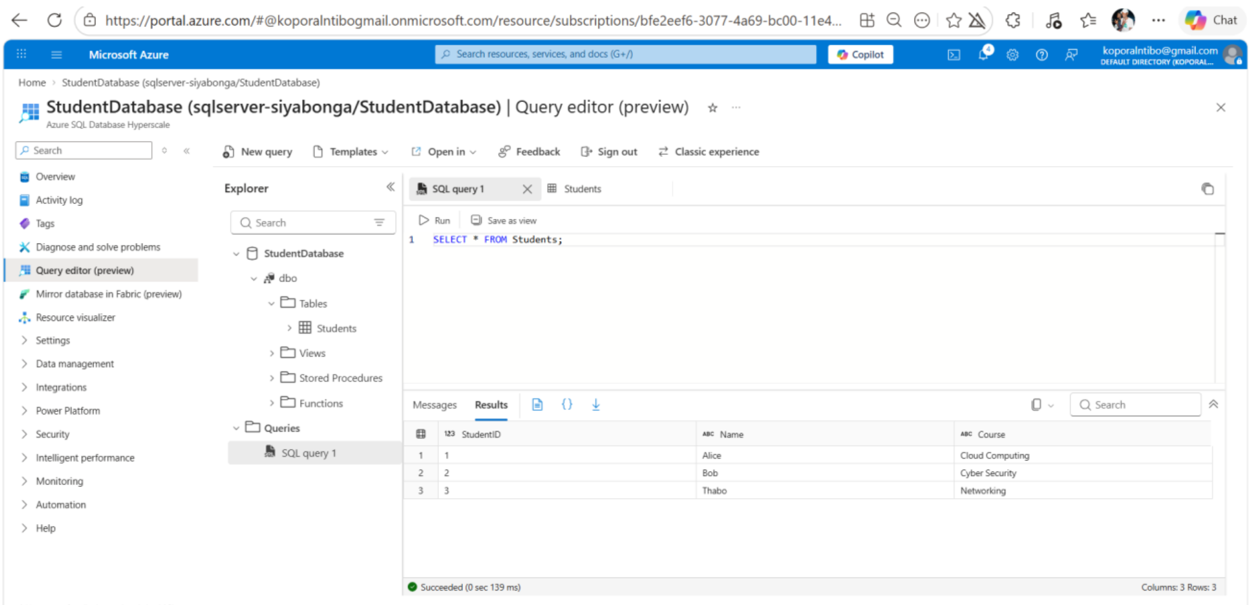
Task 3: Insert Data into Table





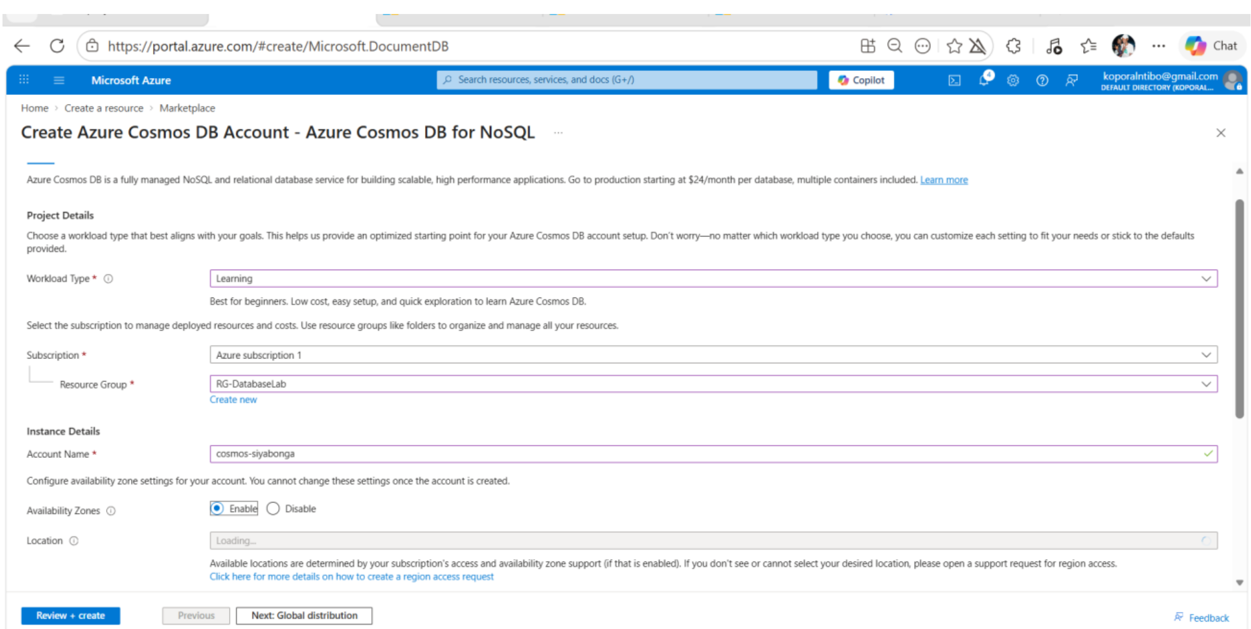
Task 4: Query Data from Database

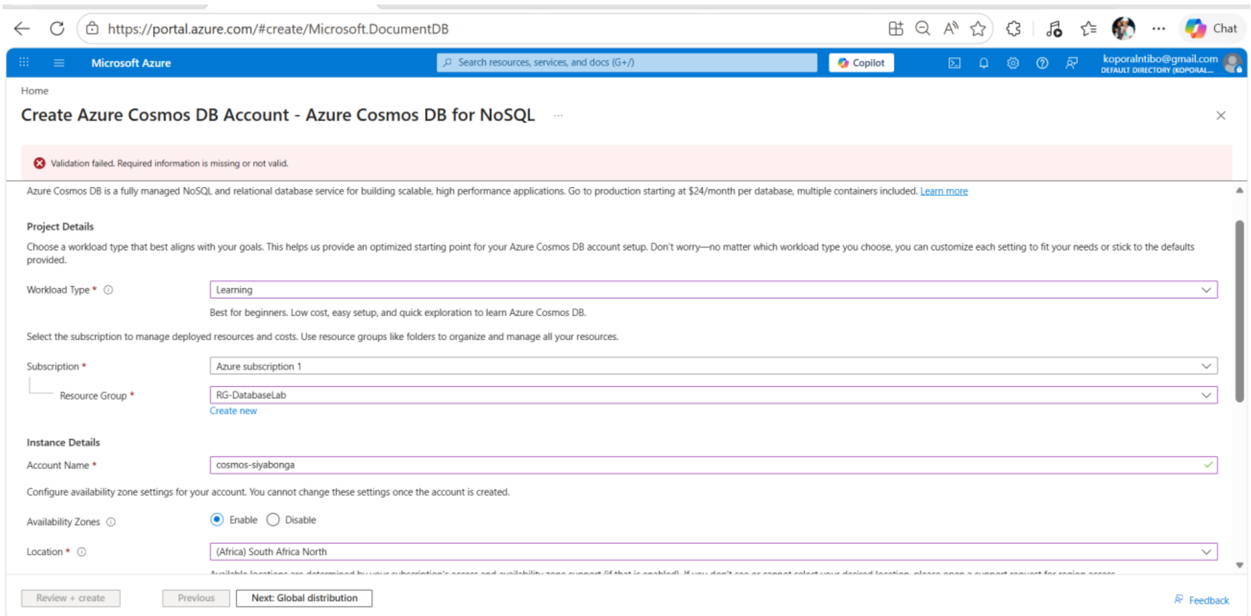
Use SELECT * FROM Students to display the full table with its’ respective data already filled in.

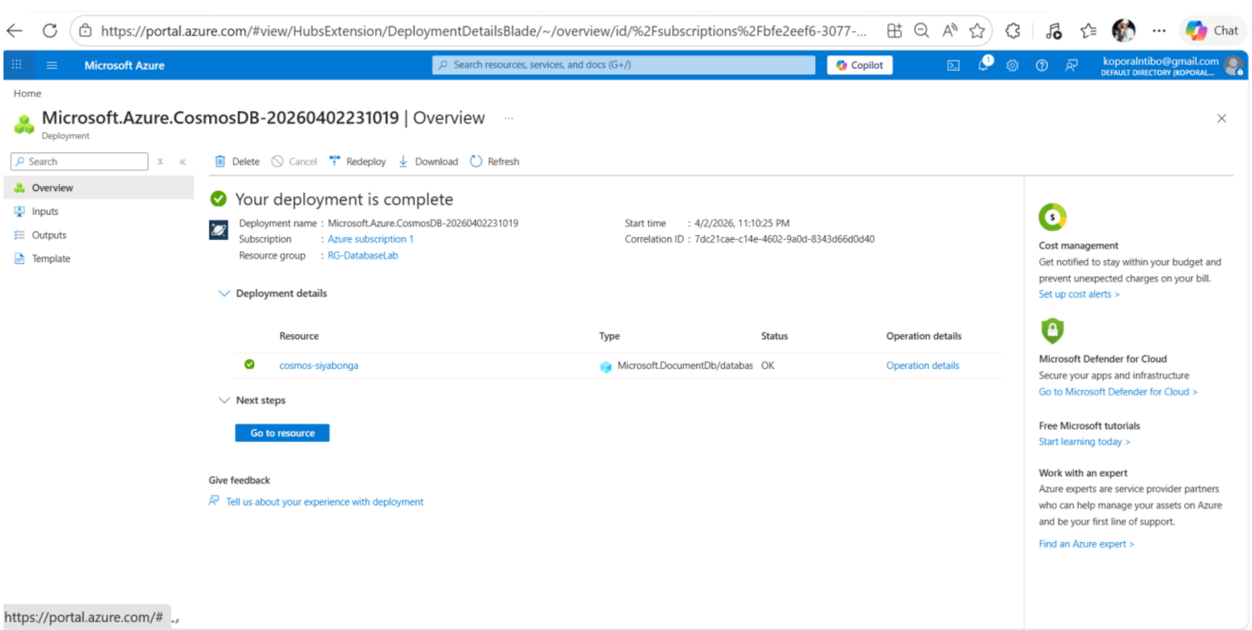


Task 5: Create Cosmo DB (NoSQL Database)

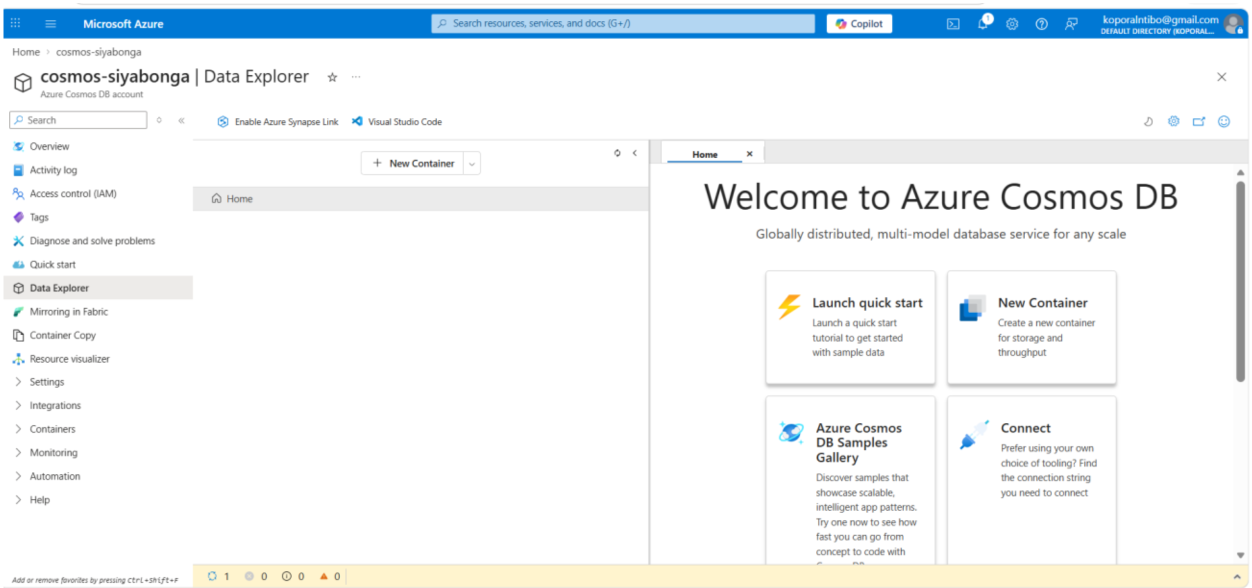
Once in the “Create Azure Cosmos DB Account – Azure Cosmos DB for NoSQL” framework, choose the learning workload as it corresponds with the subscription you have chosen which will be automatically selected, along with the resource group, due to it associated with the resource group you created before. Learning workload is best for beginners as it offers low cost, easy setup, and quick exploration to learn Azure Cosmos DB. Name the Cosmos DB account using your name to distinguish it from other potential accounts. Afterwards create the database.

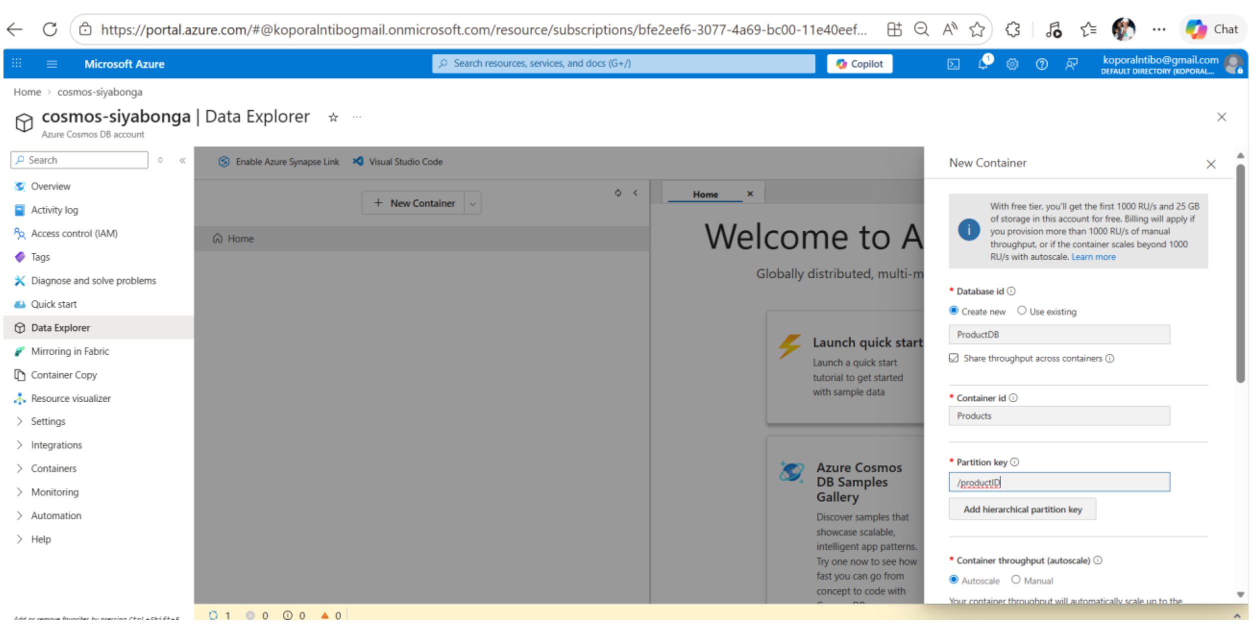


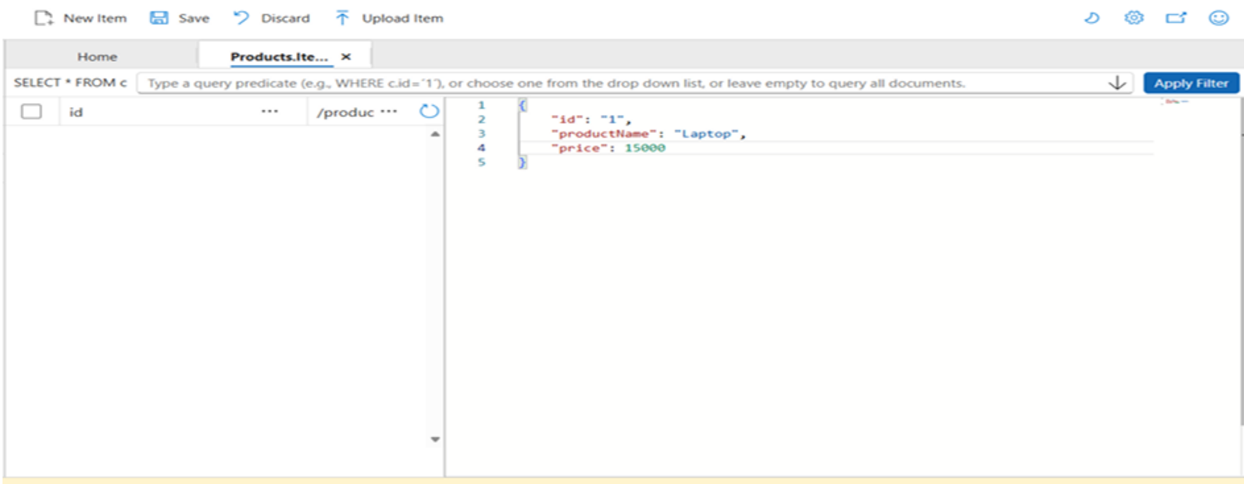

https://portal.azure.com/#create/Microsoft.DocumentDB
Chat
Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
koporalntibo@gmail.com
DEFAULT DIRECTORY (KOPORAL...
Home
Create Azure Cosmos DB Account - Azure Cosmos DB for NoSQL
Validation failed. Required information is missing or not valid.
Azure Cosmos DB is a fully managed NoSQL and relational database service for building scalable, high performance applications. Go to production starting at $24/month per database, multiple containers included. Learn more
Project Details
Choose a workload type that best aligns with your goals. This helps us provide an optimized starting point for your Azure Cosmos DB account setup. Don't worry—no matter which workload type you choose, you can customize each setting to fit your needs or stick to the defaults provided.
Workload Type *
Learning
Best for beginners. Low cost, easy setup, and quick exploration to learn Azure Cosmos DB.
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.
Subscription *
Azure subscription 1
Resource Group *
RG-DatabaseLab
Create new
Instance Details
Account Name *
cosmos-siyabonga
Configure availability zone settings for your account. You cannot change these settings once the account is created.
Availability Zones
Enable
Disable
Location *
(Africa) South Africa North
Review + create
Previous
Next: Global distribution
Feedback

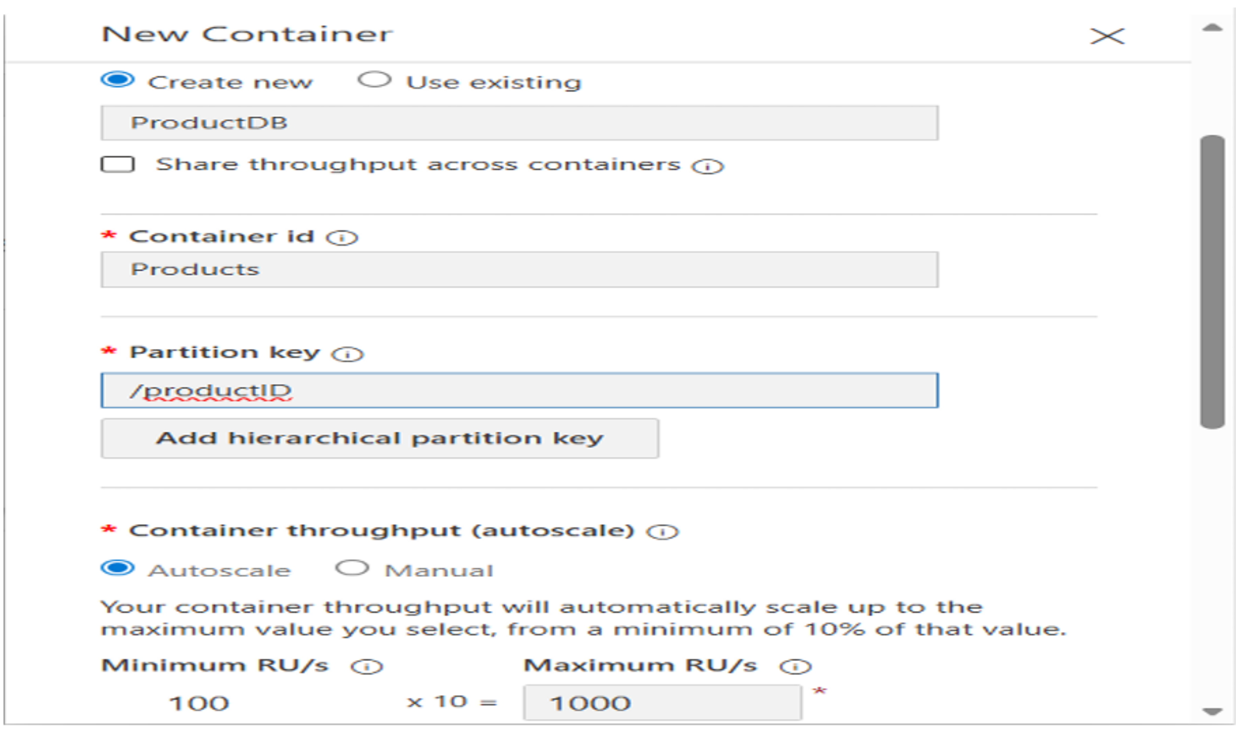

https://portal.azure.com/#view/HubsExtension/DeploymentDetailsBlade/~/overview/id/%2Fsubscriptions%2Fbfe2eef6-3077-...
Chat
Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
koporalntibo@gmail.com
DEFAULT DIRECTORY (KOPORAL...
Home
Microsoft.Azure.CosmosDB-20260402231019 | Overview
Deployment
Search
Delete
Cancel
Redeploy
Download
Refresh
Overview
Inputs
Outputs
Template
Your deployment is complete
Deployment name : Microsoft.Azure.CosmosDB-20260402231019
Subscription : Azure subscription 1
Resource group : RG-DatabaseLab
Start time : 4/2/2026, 11:10:25 PM
Correlation ID : 7dc21cae-c14e-4602-9a0d-8343d66d0d40
Deployment details
Resource
Type
Status
Operation details
cosmos-siyabonga
Microsoft.DocumentDb/databas
OK
Operation details
Next steps
Go to resource
Give feedback
Tell us about your experience with deployment
Cost management
Get notified to stay within your budget and prevent unexpected charges on your bill.
Set up cost alerts >
Microsoft Defender for Cloud
Secure your apps and infrastructure
Go to Microsoft Defender for Cloud >
Free Microsoft tutorials
Start learning today >
Work with an expert
Azure experts are service provider partners who can help manage your assets on Azure and be your first line of support.
Find an Azure expert >
https://portal.azure.com/#

TASK 6: CREATE DATABASE AND CONTAINER



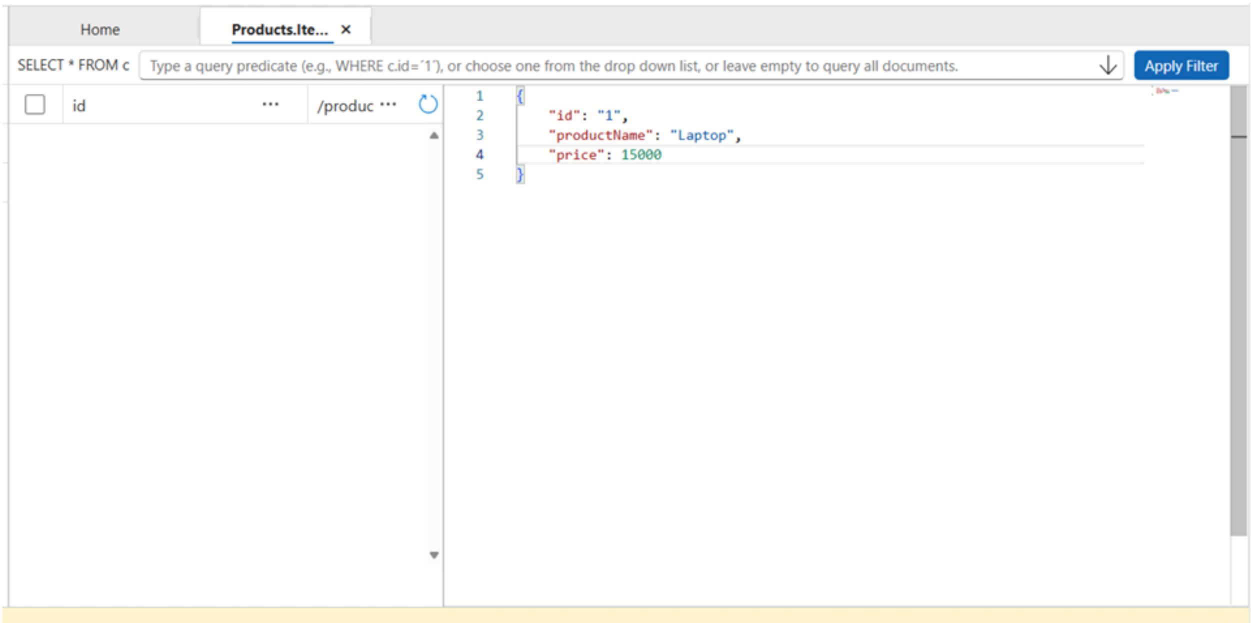


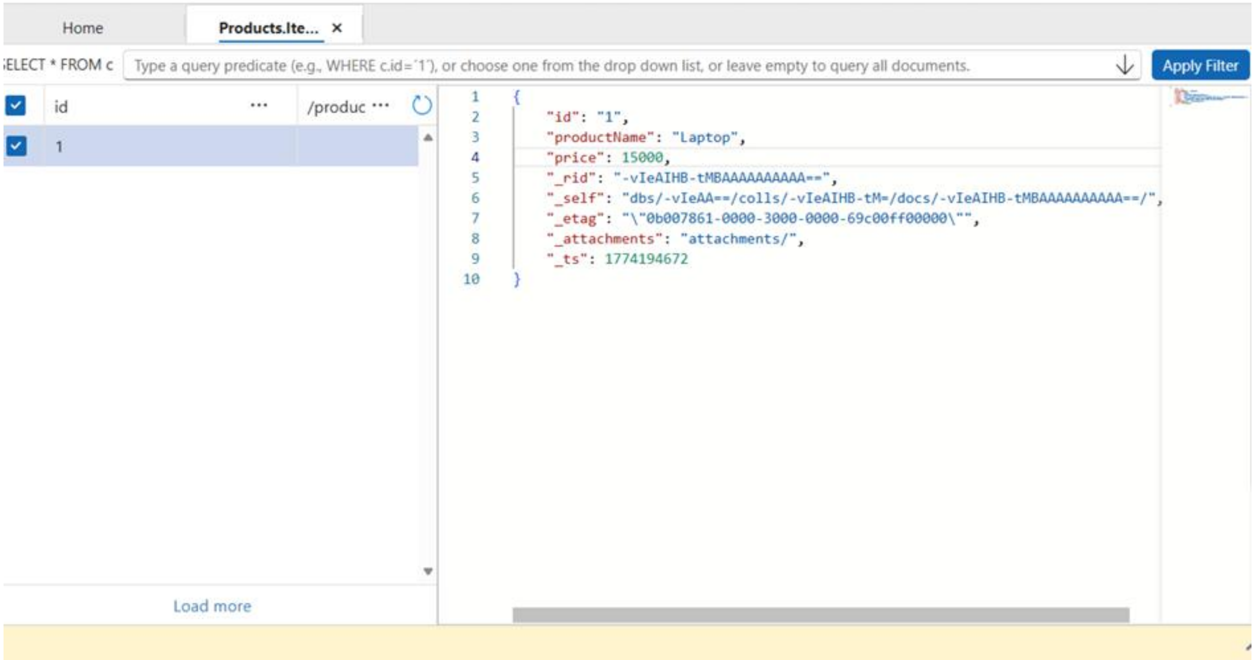

New Item
Save
Discard
Upload Item
Home
Products.Ite...
SELECT * FROM c
Type a query predicate (e.g., WHERE c.id='1'), or choose one from the drop down list, or leave empty to query all documents.
Apply Filter
id
/produc
{
"id": "1",
"productName": "Laptop",
"price": 15000
}


New Container
Create new
Use existing
ProductDB
Share throughput across containers
* Container id
Products
* Partition key
/productID
Add hierarchical partition key
* Container throughput (autoscale)
Autoscale
Manual
Your container throughput will automatically scale up to the maximum value you select, from a minimum of 10% of that value.
Minimum RU/s
Maximum RU/s
100
x 10 =
1000

Task 7: Insert JSON Document

Inserting a JSON document is the standard way of working with NoSQL databases like Azure Cosmos DB, because JSON provides a flexible, schema-less format that fits perfectly with how NoSQL stores and queries data. Use the data provided to create a new item in the newly created container.




Home
Products.Ite...
ELECT * FROM c
Type a query predicate (e.g., WHERE c.id='1'), or choose one from the drop down list, or leave empty to query all documents.
Apply Filter
id
/produc
1
{
"id": "1",
"productName": "Laptop",
"price": 15000,
"_rid": "-vIeAIHB-tMBAAAAAAAAAA==",
"_self": "dbs/-vIeAA==/colls/-vIeAIHB-tM=/docs/-vIeAIHB-tMBAAAAAAAAAA==/",
"_etag": "\"0b007861-0000-3000-0000-69c00ff00000\"",
"_attachments": "attachments/",
"_ts": 1774194672
}
Load more